

Algebra Ke Sabhi Jaruri Formulas Aur Unka Step-by-Step Derivation

Algebra (beejganit) ke formulas ko ratne ke bajay agar hum unhe samajh kar derive karein, to unhe yaad rakhna bahut aasan ho jata hai. Is article mein hum basic se lekar advanced formulas tak, har ek step ko detail mein samjhenge. Ye article un sabhi ke liye hai jo pehli baar in formulas ko deeply samajhna chahte hain.

1. Square (Varg) Wale Basic Formulas

Sabse pehle hum do terms wale square formulas ko derive karenge.

Formula 1 & 2: $(a + b)^2$

Derivation: $(a + b)^2$ ka matlab hai $(a + b)$ ko do baar multiply karna.

Ise hum ek aur tarike se likh sakte hain:

(Kyunki $(a - b)^2 = a^2 - 2ab + b^2$, isme $4ab$ jodne par waapas $+2ab$ aa jayega.)

Formula 3 & 4: $(a - b)^2$

Derivation:

Iska dusra roop:

Formula 5: $a^2 - b^2$

Ise derive karne ke liye simply in do terms ko multiply karein:

Toh formula banta hai:

Formula 6, 7 & 8: $a^2 + b^2$

Upar diye gaye pehle do formulas se hum ise nikal sakte hain:

- **Roop 1:** $(a + b)^2 = a^2 + 2ab + b^2 \rightarrow a^2 + b^2 = (a + b)^2 - 2ab$
- **Roop 2:** $(a - b)^2 = a^2 - 2ab + b^2 \rightarrow a^2 + b^2 = (a - b)^2 + 2ab$
- **Roop 3:** Agar dono ko jod dein toh: $a^2 + b^2 = \frac{1}{2} [(a + b)^2 + (a - b)^2]$

2. Teen Terms Wale Square Formulas

Formula 9 & 10: $(a + b + c)^2$

Derivation:

Pehle 'a' se, fir 'b' se, aur fir 'c' se multiply karein:

Jodne par:

Common lene par iska dusra roop banta hai:

Formula 11: $(a + b - c)^2$

Isme bas 'c' ki jagah '-c' rakhna hai:

Formula 12: $(a - b - c)^2$

Isme 'b' aur 'c' dono minus mein hain:

3. Cube (Ghan) Wale Basic Formulas

Formula 13 & 14: $(a + b)^3$

Derivation:

Multiply karne par:

Agar hum bich wale terms se $3ab$ common le lein toh:

Formula 15 & 16: $(a - b)^3$

Derivation:

Solve karne par:

Isme se $-3ab$ common lene par:

4. $a^3 + b^3$ Aur $a^3 - b^3$ Ke Formulas

Formula 17, 18, 19 & 20: $a^3 + b^3$ (4 Roop)

Derivation: Hum jante hain ki $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$. $a^3 + b^3$ ko ek side rakhein:

Roop 1: $a^3 + b^3 = (a + b)^3 - 3ab(a + b)$ **Roop 2:** Ise expand karke likhein: $(a + b)^3 - 3a^2b - 3ab^2$

Ab Roop 1 mein se dono terms mein se $(a + b)$ common le lein: **Roop 3:** $= (a + b) [(a + b)^2 - 3ab]$

Andar wale $(a + b)^2$ ko solve karein $(a^2 + b^2 + 2ab)$: $= (a + b) [a^2 + b^2 + 2ab - 3ab]$ **Roop 4:** $= (a + b) (a^2 - ab + b^2)$

Formula 21, 22, 23 & 24: $a^3 - b^3$ (4 Roop)

Derivation: Hum jante hain ki $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$. $a^3 - b^3$ ko ek side rakhein:

Roop 1: $a^3 - b^3 = (a - b)^3 + 3ab(a - b)$ **Roop 2:** Ise expand karke likhein: $(a - b)^3 + 3a^2b - 3ab^2$

Ab Roop 1 mein se $(a - b)$ common le lein: **Roop 3:** $= (a - b) [(a - b)^2 + 3ab]$

Andar wale $(a - b)^2$ ko solve karein $(a^2 - 2ab + b^2)$: $= (a - b) [a^2 - 2ab + b^2 + 3ab]$ **Roop 4:** $= (a - b) (a^2 + ab + b^2)$

5. $(a + b + c)^3$ Ka Vistar

Formula 25: $(a + b + c)^3$

Derivation:

Pehle $(a + b + c)^2$ ka man rakhein:

Isse aage solve (multiply) karne par hume milta hai:

Sabko ek sath jodne par:

Isse common lekar arrange karne par final formula banta hai:

6. Sabse Bada Aur Important: $a^3 + b^3 + c^3 - 3abc$ Ke 5 Formulas

Yeh formula kahan se aaya, bahut kam log jante hain. Aaiye iska step-by-step post-mortem karte hain.

Derivation Shuruwati Kadam: Hum jante hain ki: $a^3 + b^3 = (a + b)^3 - 3ab(a + b)$ Is value ko main expression mein rakhein:

Arrange karein:

Maan lijiye: $(a + b) = x$ aur $c = y$. Toh pehla hissa $x^3 + y^3$ ban jayega. $x^3 + y^3$ ka formula hota hai: $(x + y)(x^2 - xy + y^2)$ Ab x aur y ka value wapase rakhne par:

Yahan dono terms mein $(a + b + c)$ hai. Isse common le lein:

Andar wale $(a + b)^2$ ko solve karein:

Yahan $+2ab$ aur $-3ab$ milkar $-ab$ ban jayenge. Isse hamara **Pehla Formula** banta hai:

Formula 26: Pehla Roop

Formula 27: Dusra Roop (Add & Subtract method)

Formula 26 mein $2(ab + bc + ca)$ jodein aur ghatayen:

Shuruwati hissa $(a + b + c)^2$ ban jayega:

Formula 28: Teesra Roop (Divide & Multiply by 2)

Formula 26 mein $1/2$ se divide karein aur andar 2 se multiply karein:

Formula 29: Chautha Roop (Terms ko todna)

Formula 28 ke andar wale terms ko is tarah todein: $(a^2 - 2ab + b^2) + (b^2 - 2bc + c^2) + (c^2 - 2ca + a^2)$ Ye formula ban jayega:

Formula 30: Panchwa Roop ($a^2 + b^2 + c^2$ jodna aur ghatana)

Formula 28 mein $a^2 + b^2 + c^2$ jodein aur ghatayen:

7. Special Case (Vishesh Sthiti)

Formula 31: Agar $a + b + c = 0$ ho

Derivation: Koi bhi upar diya gaya formula le lijiye, jaise Formula 26:

Agar hum $(a + b + c) = 0$ rakhein, toh right side mein zero se multiply hone ke karan sab kuch zero ho jayega.

Ab $-3abc$ ko right side (dahini taraf) bhej dein:

Nishkarsh: Jab bhi $a + b + c = 0$ diya ho, toh hamesha $a^3 + b^3 + c^3 = 3abc$ hoga.